

RESEARCH

Open Access



Demand–capacity estimation using queueing theory: application to hospital resource planning in the 2023 Türkiye earthquake

Bircan Kara¹ and Ali Utku Şahin^{2*}

Abstract

Purpose This study aims to develop and validate a mathematical model for estimating the number of emergency physicians required in post-earthquake scenarios, using actual data from the 2023 Kahramanmaraş earthquakes in Türkiye.

Methodology The methodology follows a structured five-step framework to assess earthquake impact and emergency healthcare demand. First, population impact is analyzed using USGS PAGER data to estimate exposure levels. Second, household and building stock characteristics are profiled from TurkStat, focusing on construction year and building types, which influence structural vulnerability. Third, collapse probabilities are determined through empirical fragility functions that relate earthquake intensity to building failure rates. Fourth, casualties are estimated by combining structural damage with fatality and injury ratios specific to building types. Finally, physician demand is calculated using the M/M/s queueing theory model, incorporating key variables, such as emergency healthcare capacity, patient arrival rates, and examination duration, to forecast medical staffing needs in the aftermath of a disaster.

Findings The model estimated 11,645 potential emergency department visits in Hatay province within the first 144 h post-earthquake. Based on this, 27 emergency physicians per shift—or 81 total physicians across three shifts—are required to operate at full capacity. These figures closely align with actual post-disaster hospital admission data, validating the model.

Conclusions This study presents a scalable, actual-data-validated model for physician workforce planning in disaster scenarios. The queuing-based approach supports strategic resource allocation and enhances organizational resilience. Unlike existing models, this study directly integrates field-specific damage and population data to forecast real-time health system needs.

Keywords Disaster preparedness, Emergency medicine, Queueing theory, Human resource planning, Earthquake resilience, Mathematical modeling, Health systems

*Correspondence:

Ali Utku Şahin

aliutkusahin@mku.edu.tr

Full list of author information is available at the end of the article



© The Author(s) 2025. **Open Access** This article is licensed under a Creative Commons Attribution-NonCommercial-NoDerivatives 4.0 International License, which permits any non-commercial use, sharing, distribution and reproduction in any medium or format, as long as you give appropriate credit to the original author(s) and the source, provide a link to the Creative Commons licence, and indicate if you modified the licensed material. You do not have permission under this licence to share adapted material derived from this article or parts of it. The images or other third party material in this article are included in the article's Creative Commons licence, unless indicated otherwise in a credit line to the material. If material is not included in the article's Creative Commons licence and your intended use is not permitted by statutory regulation or exceeds the permitted use, you will need to obtain permission directly from the copyright holder. To view a copy of this licence, visit <http://creativecommons.org/licenses/by-nc-nd/4.0/>.

Introduction

Disasters are extraordinary events that severely disrupt daily activities and negatively impact human health and life [29]. According to the World Health Organization, disasters are events that can lead to large-scale loss of life and property, and their effects cannot be controlled even if they are anticipated [45]. The immediate consequences of disasters include physical injuries, trauma, and exacerbation of chronic diseases, while long-term impacts may include psychological distress and socio-economic challenges [1, 5, 9, 15]. Among various types of disasters, natural events such as earthquakes often present a higher risk for causing significant health disruptions [34, 49].

The importance of disaster-related health service planning has been demonstrated in numerous past incidents. For instance, during the 2010 Haiti earthquake, the lack of prepared healthcare services exacerbated the disaster's effects, leading to prolonged suffering and poor health outcomes [39]. Similarly, the COVID-19 pandemic underscored the critical need for health systems to adapt rapidly to large-scale emergencies, with shortages in healthcare workers and resources being a key limitation in effective response [26, 50]. In West Africa, the Ebola outbreak further illustrated how inadequate disaster preparedness in healthcare systems can result in the rapid spread of diseases, overwhelming hospitals and public health networks [40]. These examples emphasize the critical need for robust disaster preparedness models, particularly in workforce planning, as existing frameworks often fall short in addressing the complexities of human resource allocation during large-scale emergencies. Strategic workforce planning is essential to ensuring the resilience of healthcare services and optimizing resource use during such crises [14, 27, 30].

The aftermath of earthquakes not only affects individuals but also damages critical service infrastructure, such as health systems, transportation, and communication networks, which severely hampers the continuity of healthcare services [31]. Disasters like earthquakes disrupt the normal functioning of health systems, overloading emergency departments and healthcare personnel, and highlighting the urgent need for pre-disaster health service planning and preparedness [2, 20].

Studies show that during large-scale disasters, healthcare needs rise dramatically, involving both urgent clinical issues such as traumatic injuries and long-term challenges such as psychological trauma and chronic condition exacerbations [35, 45]. Health facilities often face severe operational bottlenecks due to insufficient workforce surge capacity, which has been repeatedly demonstrated in real-time trauma registries [3, 8]. Furthermore, the 2023 Türkiye earthquakes exposed gaps in triage and patient prioritization, emphasizing the critical

need for coordinated surge planning in hospital disaster preparedness [48].

Addressing these challenges requires proactive workforce planning, not just in terms of numbers, but also in skill distribution. Flexible surge capacity, which incorporates public health, education, and healthcare resource management, has been proposed as a sustainable approach to enhancing system resilience [25]. In the context of disaster resilience, healthcare workforce planning is crucial for managing both acute and chronic care needs, alongside psychological and social support [24, 18, 19].

Despite the significant importance of human resource planning, there remains a gap in the literature, especially in mathematical or operational frameworks for workforce management in disaster settings [17].

This study aims to fill an important gap in the literature by providing a strategic perspective to ensure the continuity of healthcare services during crises. In this context, we utilize a numerical demand–capacity expression derived from existing queueing theory models to estimate the required human resource capacity based on expected patient demand. The primary objective of this study is to offer a practical and scientifically sound method that can serve as a decision support tool for healthcare administrators in workforce planning.

Methodology and mathematical modeling

This study was designed to determine the number of physicians required to work in the emergency department of Hatay Mustafa Kemal University Health Practice and Research Hospital, which is the only hospital that continued to provide health service in Hatay province after the February 6th, 2023 Kahramanmaraş-centered earthquakes. In the study, a mathematical modeling-based planning method was used to determine the number of physicians needed based on the estimated number of injured.

The study was conducted by considering various variables related to the estimated number of disaster victims, emergency service capacity, examination times and working hours of health workers. Similar studies in the literature, post-disaster health services data, statistical information published by official institutions and field reports were taken as reference in the construction of the model.

The study addresses human resource planning of emergency physicians to ensure continuity of health service delivery after a disaster. The model developed in this direction is based on a methodological approach consisting of four basic steps. These steps are detailed below, respectively.

The model aims to reveal the need for physicians at a level that can meet the sudden patient load directed to the health system in the first critical hours of the disaster. The results obtained are intended to provide a scientific basis for decision makers and health managers in human resource planning in similar disaster situations.

The methodology used in this paper is shown in Fig. 1 and explained below.

Collection of data on exposure and building stock profile

The number of people affected by the Kahramanmaraş Earthquakes occurred on February 6th, 2023 and their distribution according to the shaking intensity measured in Modified Mercalli Index (MMI) was retrieved from the U.S. Geological Survey Prompt Assessment of Global Earthquakes for Response (USGS [42]).

The building and household data for the provinces affected by the earthquakes were obtained from the statistics of the Turkish Statistical Institute (TurkStat) for the year 2023 (TurkStat, 2023). The data collected in this context are as follows:

- Population of provinces in the earthquake-affected region in 2023,
- Distribution of buildings in the earthquake-affected region by year of construction at the provincial scale,
- Average number of storeys of residential buildings in the earthquake-affected area, according to the provincial scale,
- Average household sizes in provinces in the earthquake-affected region,

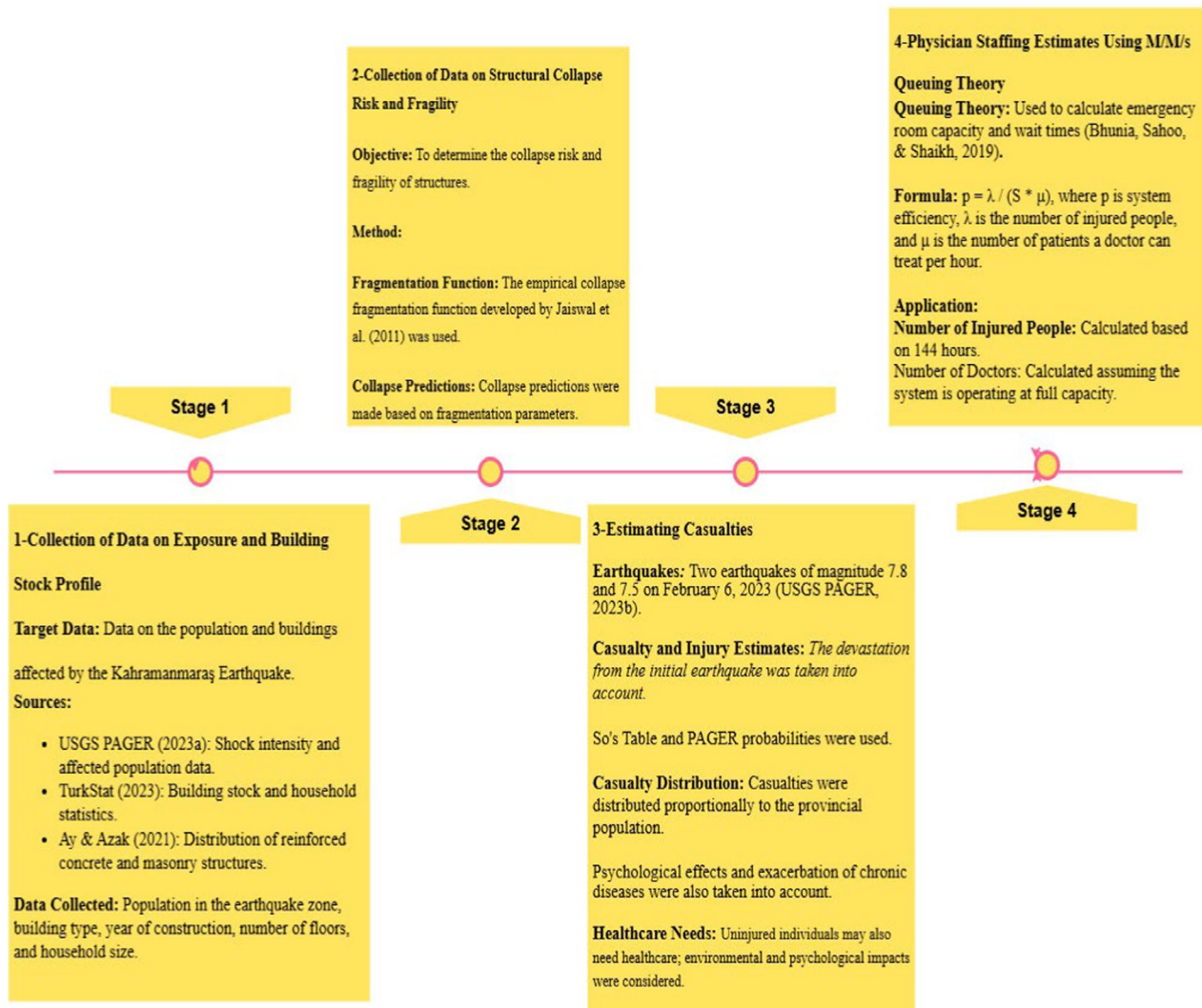


Fig. 1 Methodological framework (Authors' own work)

- v. Provincial distribution of households residing in the earthquake-affected region according to the year of construction of the structures they reside in.

In addition, the distribution of reinforced concrete and masonry structures across Türkiye by year, which could not be obtained from TurkStat data, was obtained from Ay and Eroğlu Azak [4].

With the data collected at this stage, the distribution of the buildings in which households reside according to the building material, year of construction and average floor height was reached.

Collection of data on structural collapse risk and fragility

The empirical collapse fragility function developed by Jaiswal et al. [21] was used in the collapse estimation. Accordingly, building collapse estimates were calculated by the formula shown in Formula 1, taking into account the fragility parameters according to the building type shown in Table 1.

$$F(x) = A_j * 10^{\left(\frac{B_j}{x_j - C_j}\right)}$$

Formula 1. Collapse fragility function

where A_j , B_j , and C_j are scale, shape, and location parameters of the distribution function to be estimated by fitting the prior expert judgment of experts for building type j , and x is the shaking intensity [21].

Estimating casualties

The fatality rates used in this estimation were derived based on ground shaking data from the first earthquake, which was measured using the Modified Mercalli

Intensity (MMI) scale. The MMI values were obtained from the USGS PAGER database (USGS [43]), and these were used in conjunction with So's table of casualty estimates (Table 2) and PAGER probabilities.

As it is known, the earthquakes that occurred on February 6th, 2023, consisted of two separate earthquakes with magnitudes of 7.8 and 7.5 on the Richter scale, occurring approximately 9 h apart (USGS [43]). At this stage of the study, the number of dead and injured was estimated by considering only the destruction that occurred in the first earthquake.

In the estimation of the number of dead and wounded, the So's table of casualty estimates shown in Table 2 and PAGER probabilities were used. Initially, the number of wounded was distributed proportionally to the provincial populations of the earthquake zone, with the assumption that the entire population, excluding the deceased, had been wounded and that all wounded individuals were admitted to the hospital. However, in addition to the direct injuries caused by the earthquake, we acknowledge that individuals who were not physically injured during the initial tremors may still experience significant health-related needs. In large-scale disasters, even individuals in undamaged buildings may suffer from secondary effects such as physical injuries, psychological trauma, or exacerbation of pre-existing conditions due to aftershocks, disruptions in daily life, or the breakdown of healthcare systems [16, 31]. Furthermore, those with chronic conditions, such as hypertension, diabetes, or respiratory diseases, may face exacerbations of their health conditions in the absence of regular medical care [7].

In addition, the psychosocial impact of such events can lead to the emergence of mental health issues, including post-traumatic stress disorder (PTSD), anxiety, and depression, which may necessitate healthcare interventions [16]. The disruption of basic living conditions, including shelter and sanitation, increases vulnerability, especially in areas with limited infrastructure or overcrowded emergency shelters [7]. These conditions contribute to an increased need for healthcare services, even for individuals who are not physically injured.

Therefore, our model includes the assumption that individuals who were not initially injured could still require medical attention, particularly in the form of chronic disease management and mental health support.

Table 1 Collapse fragility parameters for the building types that are affected by the earthquake [21]

Building type	A	B	C	R ²
Nonductile concrete moment frame (C3)	3.42	−5.03	5.62	0.99
Brick masonry with lime/cement mortar (UFB2)	8.03	−7.59	4.60	0.99

Table 2 Judgment-based fatality rates for the building types that are affected by the earthquake [38]

Building classes	PAGERSTR	Typical volume loss (%)	Fatality ranges from literature (%)	Suggested fatality rate (for pager) (%)
Irregular stone low-rise concrete slab roofs	RS5	> 70	10–40	30.0
Concrete frame—mid-rise	C3 M	> 20	10–25	25.0

This leads to the inclusion of the entire population (excluding fatalities) in the estimation, reflecting the broader health needs during such crises. In summary, the healthcare demand in the aftermath of large-scale disasters is not limited to physical injuries but extends to the exacerbation of pre-existing conditions and the emergence of new health needs due to environmental, psychological, and social stressors [15, 31].

Physician staffing estimates using M/M/s queuing theory

Queueing theory, a method within operations research, is used to analyze the service process with random variation in demand and service times. The purpose of queueing theory analysis is to calculate the level of service that needs to be provided, in other words, the required service delivery capacity [6]. With this analysis, the waiting and service times in a “queue” can be revealed [33]. The basic elements of queueing theory are divided into service beneficiaries and service providers. The service beneficiaries are represented on an hourly scale according to their arrival time to receive the service, while the service itself is measured by the capacity of the service providers, also on an hourly scale [33].

In this context, a relationship can be established between the hourly average of service applicants (λ) and the average service delivery time (μ) of a service provider (S) in terms of system efficiency (ρ) as shown below (Formula 2) [33].

$$\rho = \lambda / (S * \mu)$$

Formula 2. System utilization factor

The equation shown in Formula 2 can also be used to calculate the number of physicians that should be assigned to emergency services after an earthquake. When one matches the variables of the queueing theory with the case; service applicants could be interpreted as the patients admitting to the hospital, and service providers (or channels) could be interpreted as the physicians. As is known, p represents the capacity of the system, and, in this case, which is an emergency healthcare service right after an earthquake, should be taken as 1 to represent that the capacity of the hospital is 100%. In this case, λ is the hourly average of the estimated number of injured people who are assumed to be admitted to the hospital, and μ is the number of patients/injured that a physician can deal with in 1 h.

According to the literature, the maximum time required to be rescued alive from a collapsed building is 144 h [36], and an average of 20 min is allocated for the emergency room examination of a patient in the USA and the UK [37, 47].

Considering these data, the hourly average of the estimated number of injured people who are assumed to be

admitted to the hospital (λ) was calculated over 144 h, μ was calculated over 3 injured/hour, and the number of physicians that should be assigned to the emergency department was calculated using the M/M/s type queueing theory model according to the $\rho=1$ situation in which the hospital emergency department will operate at full capacity after an earthquake.

Findings

The following section summarizes the results of a four-stage demand–capacity estimation model designed to determine emergency physician requirements after the 2023 Kahramanmaraş earthquakes. Key estimates include the total number of casualties, distribution of affected households by building type and severity, and physician staffing needs derived using M/M/s queueing theory. Where possible, model estimates are validated against real hospital admission data from Hatay province.

Exposure and building stock profile

More than 20 million individuals were exposed to seismic intensities of MMI 6 or greater. Among them, over 2 million were in areas of MMI 8+, where structural vulnerability is highest. In total, nearly 49% of all households in the affected provinces resided in buildings constructed before 2000, with Hatay showing a balanced distribution between pre- and post-2000 construction.

The dominant building types included reinforced concrete frames (C3M) and unreinforced brick masonry (UFB), both of which exhibit distinct fragility profiles under seismic stress.

Table 3 shows the number of people affected by the Kahramanmaraş Earthquakes that occurred on February 6, 2023, and their distribution according to the Mercalli severity scale (MMI).

The number and proportion of households by building construction year, along with the average household sizes and the average number of storeys in the eleven provinces affected by the earthquakes that occurred on February 6th, 2023, are shown in Tables 4 and 5.

The populations of the provinces in the earthquake region and their ratio to the regional population are shown in Table 6.

Table 3 Affected population according to MMI (USGS [44])

MMI	Number of affected people
6	13,482,000
7	6,354,000
8	1,928,000
9	746,000

Table 4 Distribution of number of households by year of building construction in earthquake region provinces [41]

Province	Number of households by year of building construction				Total number of household
	Before 1980	1980–2000	After 2001	Unknown	
Adana	82,344.00	220,496.00	244,668.00	85,367.00	632,875.00
Adıyaman	13,547.00	36,586.00	81,271.00	23,896.00	155,300.00
Diyarbakır	25,818.00	104,960.00	229,532.00	34,558.00	394,867.00
Elazığ	17,396.00	41,103.00	91,777.00	23,560.00	173,836.00
Gaziantep	34,351.00	135,511.00	269,876.00	83,209.00	522,947.00
Hatay	60,710.00	146,509.00	224,464.00	17,468.00	449,151.00
Kahramanmaraş	36,479.00	83,717.00	181,016.00	10,247.00	311,458.00
Kilis	4464.00	8680.00	20,917.00	5959.00	40,020.00
Malatya	32,239.00	64,869.00	111,603.00	21,787.00	230,499.00
Osmaniye	16,465.00	40,070.00	72,686.00	26,979.00	156,199.00
Şanlıurfa	22,769.00	76,306.00	251,163.00	61,183.00	411,421.00
TOTAL	346,582.00	958,807.00	1,778,973.00	394,213.00	3,478,573.00

Table 5 Household sizes, building construction year, and storey distribution in earthquake region provinces [authors' own work according to [41]]

Province	Total number of households	Number and ratio of households by years of building construction				Average household size	Average number of storeys
		Number of households (before 2000)	Ratio of households (before 2000)	Number of households (after 2001)	Ratio of households (after 2001)		
Adana	632,875.00	388,207.00	0.6134	244,668.00	0.3866	3.30	5.43
Adıyaman	155,300.00	74,029.00	0.4767	81,271.00	0.5233	3.70	3.54
Diyarbakır	394,867.00	165,336.00	0.4187	229,532.00	0.5813	4.20	5.88
Elazığ	173,836.00	82,059.00	0.4720	91,777.00	0.5280	3.10	4.34
Gaziantep	522,947.00	253,071.00	0.4839	269,876.00	0.5161	3.80	4.96
Hatay	449,151.00	224,687.00	0.5002	224,464.00	0.4998	3.30	3.59
Kahramanmaraş	311,458.00	130,443.00	0.4188	181,016.00	0.5812	3.40	4.36
Kilis	40,020.00	19,103.00	0.4773	20,917.00	0.5227	3.30	3.65
Malatya	230,499.00	118,895.00	0.5158	111,603.00	0.4842	3.10	4.95
Osmaniye	156,199.00	83,514.00	0.5347	72,686.00	0.4653	3.30	3.29
Şanlıurfa	411,421.00	160,258.00	0.3895	251,163.00	0.6105	4.70	3.71
TOTAL	3,478,573.00	1,699,602.00	0.4886	1,778,973.00	0.51	*3.56	*4.33

* MEDIAN

Using the data in Tables 5 and 6, the household distribution of the data shown in Table 3 by severity and year was made and shown in Table 7.

According to Ay and Eroğlu Azak, the proportion of masonry and reinforced concrete structures built before 2000 was 44.43% and 41.29%, respectively, while these rates were 5.19% and 79.29% for post-2000 structures [4]. Considering these rates, which cover 85% of the residential buildings in Türkiye, the distribution of households by masonry and reinforced concrete structures at the provincial scale is shown in Table 8. The distribution

shown in Table 8 is based on the UFB codes corresponding to masonry buildings and C3 codes corresponding to reinforced concrete buildings among the building types in Table 1.

Structural collapse risk and fragility

Using the empirical fragility functions from Jaiswal et al. [21], collapse probabilities were calculated for each building type across different intensity levels (MMI). The results show that UFB buildings constructed before 2000 have a collapse probability of

Table 6 Population of the earthquake region provinces and their ratios to the regional population [41]

Province	Population	Ratios to the regional population
Adana	2,280,484.00	0.1638
Adıyaman	611,037.00	0.0439
Diyarbakır	1,833,684.00	0.1317
Elazığ	603,941.00	0.0434
Gaziantep	2,193,363.00	0.1575
Hatay	1,562,185.00	0.1122
Kahramanmaraş	1,134,105.00	0.0814
Kilis	156,739.00	0.0113
Malatya	750,491.00	0.0539
Osmaniye	561,061.00	0.0403
Şanlıurfa	2,237,745.00	0.1607
TOTAL	13,924,835.00	

15.1% at MMI 9, while C3M buildings, although at a lower probability, still exhibit significant collapse risks.

These collapse probabilities were then combined with household distribution and average occupancy data. This analysis led to an estimation of approximately 39,000 deaths and 104,000 injuries across the affected region.

Using Formula 1, the collapse probabilities for each building type and construction year, along with the corresponding number of households in these buildings, were calculated. The results are shown in Table 9. In addition, taking into account the average storey height

from Table 5, C3 buildings were evaluated as C3M structures.

According to these data, the total number of households residing in potentially collapsed buildings by building type and year of construction is shown in Table 10.

According to the probability of death according to the structure types shown in Table 2, the probability of death in the C3M type structure is 0.25. Since the UFB type structure is not included in Table 2, the value of 0.30 for the RS5 type structure, which shows a similar feature, is taken into consideration. As stated in the methodology and mathematical modeling section of the study, since it is assumed that all individuals outside the probability of death will be injured, the probability of injury for the C3M type structure is 0.75, and the probability of injury for the UFB type structure is 0.70. According to the calculation made by taking into account the average household size shown in Table 6, it was found that the estimated death toll for the entire earthquake zone is 38,478.75, and the number of injured is 103,800.10.

Estimated casualties in Hatay

Since the main purpose of this study is to calculate the number of physicians that should be assigned to emergency department of Hatay Mustafa Kemal University Health Practice and Research Hospital using queuing theory, the number of injured people obtained was distributed proportionally for Hatay province by taking into account the ratios shown in Table 6 and the number of injured people who are likely to apply to the hospital emergency department in Hatay province was calculated as 11,645.

Table 7 Distribution of households by year of construction and severity (authors' own work.)

MMI	Number of households living in buildings constructed before 2000	Number of households living in buildings constructed after 2000
9	102,298.46	107,075.78
8	264,385.31	276,732.04
7	871,319.63	912,010.05
6	1,848,777.35	1,935,114.30

Table 8 Distribution of households by severity (MMI), year and building category (authors' own work)

Year and building category	Ratio	MMI 9	MMI 8	MMI 7	MMI 6
UFB (before 2000)	0.4443	45,451.2080	117,466.3927	387,127.3129	821,411.7773
C3 (before 2000)	0.4129	42,239.0362	109,164.6940	359,767.8765	763,360.1685
UFB (after 2000)	0.0519	5,557.2329	14,362.3928	47,333.3215	100,432.4583
C3 (after 2000)	0.7929	84,900.3847	219,420.8333	723,132.7670	1,534,352.5283

Table 9 Probability of collapse of structures and number of households by construction year and structure type (authors' own work)

Year and building category	Probability of collapse			
	MMI 9	MMI 8	MMI 7	MMI 6
UFB before 2000	0.15125702	0.04702866	0.00552352	0.00003043
UFB after 2000	0.15125702	0.04702866	0.00552352	0.00003043
C3M before 2000	0.11113745	0.02633788	0.00077464	0.00000000
C3M after 2000	0.11113745	0.02633788	0.00077464	0.00000000
Number of households	MMI 9	MMI 8	MMI 7	MMI 6
UFB before 2000	6874.81	5524.29	2138.30	24.99
UFB after 2000	840.57	675.44	261.45	3.06
C3M before 2000	4694.34	2875.17	278.69	0.00
C3M after 2000	9435.61	5779.08	560.17	0.00

Table 10 Total number of households residing in buildings likely to collapse by building type and year of construction (authors' own work)

	Before 2000	After 2000
C3M	7848.19	15,774.86
UFB	14,562.40	1780.52
TOTAL	22,410.59	17,555.37

When these data obtained as a result of the calculations are compared with the actual data announced after the Kahramanmaraş Earthquakes in 2023, they corresponds with the number of injured announced in the entire earthquake region and the actual number of hospital admissions in Hatay province. As a matter of fact, according to EM-DAT data, the death toll in the Kahramanmaraş Earthquakes was 53,000 and the number of injured was 107,000 [13]. In addition, according to the data obtained from the hospital information management system (HIMS) of Hatay Mustafa Kemal University Health Application and Research Hospital, the number of patients admitted to the emergency department in the first 144 h after the earthquake was 8519.

Physician staffing estimates using M/M/s queueing theory

Considering the number of injured people estimated, λ shown in Formula 2 was calculated as 80.8630 ($\lambda = 80.8630$); μ was calculated as 3 ($\mu = 3$); when these values were substituted in Formula 2, S was found as 26.9560 ($S \cong 27$) for the case $\rho = 1$.

As a result of the calculations, the number of emergency physicians required for the full examination of 11,645 injured people who are predicted to be admitted to the hospital in 144 h was found to be 27. However, this calculation is based on the assumption that the hospital

will operate at full capacity and that the emergency physicians on duty will work continuously for 144 h. Therefore, the number of 27 should be considered as one shift, and the number of emergency physicians should be considered as 81, considering that the 24 h are worked in three shifts (Table 11).

Summary of key findings

The proposed modeling framework successfully estimates short-term healthcare system needs following a seismic disaster. In Hatay province, model outputs closely match real hospital admission data, providing robust support for the use of queuing theory in disaster human resource planning. The ability to determine real-time staffing needs from infrastructure and population data represents a scalable and transferable approach to disaster preparedness.

Discussion

This study aims to fill an important gap in terms of assessing the capacity of the health system in times of disaster by calculating the number of physicians required to work in emergency services according to an earthquake scenario. The obtained results reveal how critical physician planning is for the sustainability of health services under similar disaster conditions. In the discussion section, these findings are evaluated in comparison with the existing literature, and implications for optimizing health service delivery are presented.

It is known that there are models based on human resource planning in health services [22]. Similar to this method, there are models based on estimating the number of physicians that Canada will need. In that study, a model proposal was put forward by focusing on the increase in cancer cases in the future [12]. Similarly, Dall et al. [11] analyzed to reveal the need for neurologists in

Table 11 Physician staffing estimates using M/M/s queueing theory based on real data and calculations for 8519 patients (authors' own work)

Variable	Meaning	Value	Meaning (needed for 8519 patient)	Physician required for 8519 patients (M/M/s queueing theory)	Number of physicians working in the ER during the earthquake (Real life data)
λ	Patient arrivals/hour	80.86	Patient arrivals/hour	59.15	59.15
μ	Physician capacity (patients/hour)	3	Physician capacity (patients/hour)	3	108
S	Physicians needed (per shift)	27	Working physician (per shift)	19	12
Shifts	Daily shifts (8 h)	3	Daily shifts (8 h)	3	3
Total physicians	Needed for 144 h	81	Working for 144 h in the emergency service	58	36

the United States and focused on predicting the need for neurologists according to the increase in the incidence of the disease in the future. Again, in a study conducted in Taiwan, the number of pediatric physicians needed in the future according to birth rates was tried to be estimated based on a model [46]. It is seen that these models and calculations can work at optimal times and are mostly based on the sustainability of the system in the future. However, in this study, the number of physicians needed in the emergency department in extraordinary situations was modeled based on actual data. While this is similar to other studies in that it plans for the future, it differs from other studies in that it predicts the resilience and sustainability of the health system in times of emergency.

Studies on human resource planning modeling for emergencies are quite limited. Although a strategy and modeling were developed during the 2020 COVID-19 pandemic, a plan was made based on the general patient needs and load, and a plan was made with 6-h shifts, with 4 patients per 1 nurse per shift, and in an outpatient service with 600 patients per day. As a result, it was found that an average of 45 nurses should work [28]. In this study, by calculating the model inputs with real numerical values, it is revealed that the number of physicians can be calculated in terms of health system resistance in extraordinary situations such as pandemics.

The transient modeling approach developed by Paul et al. [32] clearly demonstrated how hospital operations vary in terms of human resources during disasters. In that study, similarly, the number of physicians needed in different phases of the disaster was determined through modeling, and a more flexible resource planning was presented by considering the time-dependent change of operational loads [32]. The main advantage of transient modeling is the ability to measure the responsiveness of the system under sudden patient congestion. In this context, our proposed physician scheduling model provides

a scalable, field-applicable solution designed considering the same parameters (e.g., patient arrival rate, physician capacity, waiting time). In addition, the fact that this model is worked with actual data is considered to be more applicable with more precise results than other simulation models.

The emergency service metamodel developed by Kaptan [23] defines organizational resilience based on waiting time during disasters. In that study, the capacity responsiveness of post-disaster health systems is modeled similarly by focusing on the number of physicians [23]. Similar to the “emergency plan” practices suggested in Kaptan’s model for increasing organizational resilience, the model we propose suggests a rapid increase in the number of physicians. This model also focuses on calculating the number of physicians, one of the most important human resource elements of the healthcare system, to increase the resilience of the healthcare system against disasters. In this way, it is envisaged that the need for physician manpower during a disaster can be determined, and the ability of the health system to respond quickly in terms of treatment and care will increase. In addition, this model is based on a real event and can be used in practice or in the planning phase for expected disasters.

The resilience index defined in the study of Cimellaro et al. [10] is used to measure how resilient a hospital system is against sudden disasters such as earthquakes. The approach in this study is thought to make the system more resilient and flexible by determining the number of physicians needed in the first response phase after a disaster. It reveals very important metric measurements in terms of increasing the operational capacity of the system and the sustainability of the system by providing the opportunity to plan manpower or to intervene quickly. From this perspective, the physician planning model we propose overlaps with Cimellaro’s definition

of “organizational resilience” and shows that human resources should be modeled not only as an operational but also as a strategic resource.

Limitations

In this study, we encountered several limitations related to the use of fragility data and the application of the HAZUS model:

Regional differences: The fragility functions from Jaiswal, Wald, & D'Ayala (2011) were derived from data that may not fully represent the structural characteristics and building conditions in our specific region. These functions were applied cautiously, taking into account the limitations of regional data.

Data availability: While HAZUS provides a comprehensive set of data, we could not apply it in this study due to its U.S.-centric parameters. Instead, we utilized locally available MMI parameters and fragility data to ensure the validity of our estimates.

Although the importance of healthcare personnel's experience and professional competence is widely recognized in disaster response literature, these qualitative variables were not included in the current estimation model due to the lack of standardized data and quantifiable metrics. Future studies should aim to incorporate such dimensions into more comprehensive planning frameworks. In addition, this study is based on a single case study hospital located in a high seismic risk area. While the proposed model provides a replicable framework, its direct application to other healthcare facilities may be constrained due to contextual differences in hospital infrastructure, staffing profiles, and local emergency protocols.

Conclusion and recommendations

The findings of this study show that the number of physicians needed for the sustainability of health systems in extraordinary situations, such as earthquakes, especially in emergency services, can be calculated through mathematical modeling. This model, developed based on actual earthquake data, provides a more concrete assessment of the capacity limits of healthcare services in times of disasters and emphasizes once again that human resource planning of emergency physicians is a critical element that directly affects the responsiveness of the healthcare system in disasters. In addition, the data obtained from the model is validated with hospital data, and it is seen that the model provides results close to reality.

The most important difference that distinguishes our model from other models is that the data related to the earthquake that occurred is based on the results of a real earthquake announced by the USGS Pager. In other words, it is based on the relationship between the

intensity and population of an earthquake that has actually occurred, not the possible results of a hypothetical earthquake. Therefore, the estimates of destruction, number of dead and injured stated in the methodology and mathematical modeling section of the study are based on real earthquake data. As a matter of fact, the small difference between the actual number of injured people admitted to the hospital and the number of injured people estimated as a result of the model shows the validity of the model.

Unlike other studies in the literature, this model aims to address short-term but high-intensity health needs in extraordinary situations to increase organizational resilience. In line with the organizational resilience framework presented by Kaptan [23] and Cimellaro et al. [10], human resources should be considered as a strategic element. Paul et al. [32] presented a simulation by considering varying operational risks and loads with modeling approaches. This study proposes a flexible model by taking into account varying operational loads at different stages of the disaster, but it is designed with real data and proposes an effective, scalable, and less error-prone resource planning model.

As shown in Table 11, in the aftermath of the 2023 Türkiye earthquakes, during the first 144 h, only 12 physicians per shift were available in emergency department, which was significantly insufficient to meet the overwhelming demand for medical care. According to our calculations through the model, as the number of patients admitted to the emergency department in the first 144 h was 8519, then there should be 19 physicians per shift were available in emergency department. This difference created operational bottlenecks and coordination challenges, as the Ministry of Health was primarily responsible for managing the coordination of available resources. The lack of foresight in estimating the required number of physicians contributed to delays and inefficiencies in responding to the disaster.

From the experience of the 2023 Türkiye earthquakes, it became evident that the Ministry of Health could have utilized emergency response codes to call physicians to extraordinary duties, allowing for more efficient mobilization of healthcare personnel. In addition, strengthening the Ministry's coordination networks could have helped in quickly deploying the necessary workforce across the affected regions. The ability to predict and allocate the right number of physicians and healthcare workers during such crises is critical for avoiding these challenges and ensuring an efficient disaster response.

Based on the findings of this study:

- i. Use of the designed model before possible disasters, during the disaster preparedness process,

and planning phase. The model can assist in estimating the required number of healthcare workers and ensure that the healthcare system is properly staffed and prepared ahead of time.

- ii. Evaluating the number of physicians as a strategic input to increase the operational capability of the health system and conducting planning studies accordingly. By considering physician availability as part of overall disaster planning, health systems can be more adaptable and responsive.
- iii. Use of new modeling approaches validated and calibrated with real field data alongside simulation and assumption-based models. The integration of actual data ensures that resource allocation is more accurate and adaptable to real-world scenarios.
- iv. Mathematical models such as these should be used not only theoretically but also in practice as a decision support tool and integrated into exercises. Including such models in disaster response exercises can improve the health system's capacity to deploy the right number of healthcare workers promptly.
- v. Central and local policymakers should use such models as a guide for planning other disaster coordination units. These models should be part of broader disaster response strategies, helping policymakers better allocate resources and prepare for a range of potential crises.
- vi. Guiding researchers for the design of models that can be used in other disasters, not just earthquakes. The methodology presented can be adapted to other types of disasters, including floods, pandemics, or large-scale industrial accidents, ensuring preparedness across a variety of situations.

Given the limitations of the current system, calculating the number of required physicians and efficiently mobilizing them during an emergency can significantly enhance the preparedness and response capacity of the health system. This model presents a practical tool for improving healthcare staffing efficiency in disaster scenarios, ensuring that the right number of physicians is available where needed and that coordination is streamlined across the healthcare network.

Finally, while this study specifically addresses earthquakes, it is essential to consider similar models for other disaster types. This approach would provide a comprehensive, multi-hazard strategy for disaster response, ensuring healthcare systems are better prepared for any type of catastrophic event.

Author contributions

All authors have equally contributed.

Data availability

No datasets were generated or analysed during the current study.

Declarations

Competing interests

The authors declare no competing interests.

Author details

¹Hatay Mustafa Kemal University Health Application and Research Hospital, 31060 Antakya, Hatay, Türkiye. ²Faculty of Health Sciences, Hatay Mustafa Kemal University, 31060 Antakya, Hatay, Türkiye.

Received: 13 May 2025 Accepted: 21 October 2025

Published online: 26 November 2025

References

- Altun F. Afetlerde psikososyal hizmetler: marmara ve van depremleri karşılaştırmalı analizi. *İstanbul Sabahattin Zaim Üniversitesi Sosyal Bilimler Dergisi*. 2016;4(8–9):183–97.
- Alpert EA, Malkin M, Kobliner-Friedman D. If you rebuild it, they will come—The contribution of the Israel defense forces field hospital team to the treatment of the 2023 earthquake victims in Turkey. *J Emerg Manag*. 2025. <https://doi.org/10.5055/jem.0870>.
- Al-Thani H, Frazier T, Hertelendy AJ, Asim M, El-Menyar A, Consunji R. Surge capacity crisis and mitigation plan in a trauma setting based on real-time national trauma registry data. *Disaster Med Public Health Prep*. 2022;17:e263.
- Ay BÖ, Azak TE. Türkiye’de değişen yapı özelliklerinin karşılaştırmalı incelemesi. *Çukurova Üniversitesi Mühendislik Fakültesi Dergisi*. 2021;36(4):1111–26.
- Balikuddembe JK, Reinhardt JD, Vahid G, Di B. A scoping review of post-earthquake healthcare for vulnerable groups of the 2023 Turkey-Syria earthquakes. *BMC Public Health*. 2024;24(1):945.
- Bhunja AK, Sahoo L, Shaikh AA. *Advanced optimization and operations research*. Berlin: Springer; 2019.
- Bilgehan T, Akca A, Karşigil P, Dündar E, Ayaz-Alkaya S. Determination of health needs of earthquake victims with non-communicable diseases. *Disaster Med Public Health Prep*. 2025;19:e89.
- Birinci Ş. Technical efficiency of post-disaster health services interventions: the 2023 Kahramanmaraş Earthquake in Turkey. *Int J Health Management Tour*. 2023;8(2):187–203.
- Chen G, Liu Y, Ma Z, Yang J, Beer M, Chian SC, et al. Assessing extent of building damage following an earthquake: case study of the 2023 Turkey-Syria doublet. *NPJ Nat Hazards*. 2025;2(1):51.
- Cimellaro GP, Reinhorn AM, Bruneau M. Seismic resilience of a hospital system. *Struct Infrastruct Eng*. 2010;6(1–2):127–44.
- Dall TM, Storm MV, Chakrabarti R, Drogan O, Keran CM, Donofrio PD, et al. Supply and demand analysis of the current and future US neurology workforce. *Neurology*. 2013;81(5):470–8.
- Edwards JP, Datta I, Hunt JD, Stefan K, Ball CG, Dixon E, et al. A novel approach for the accurate prediction of thoracic surgery workforce requirements in Canada. *J Thorac Cardiovasc Surg*. 2014;148(1):7–12.
- EM-DAT. 2025. Retrieved March 10, 2025 <https://public.emdat.be>.
- Goniewicz K, Ratnayake AS, Khorram-Manesh A. Resilient emergency medical systems for 21st-century complex world. *Emerg Critical Care Med*. 2025;5(1):40–4.
- Guo C, Li S, Chan SSS. Long-term effects of disaster exposure on health care workers’ resilience: a comparison of the Wenchuan earthquake-exposed and unexposed groups. *Int J Disaster Risk Reduct*. 2022;67:102658.
- Gundi M, Kaikobad R, Sharma S. Diversity in approaches in community-based mental health interventions in India: a narrative review and synthesis. *Camb Prism: Glob Ment Health*. 2025;12:e89.
- Hasan MK, Nasrullah SM, Quattrocchi A, González PA. Hospital surge capacity preparedness in disasters and emergencies: a systematic review. *Public Health*. 2023;219:74–85.

18. Hick JL, Koenig KL, Barbisch D, Bey T. Surge capacity concepts for health care facilities: the CO-S-TR model for initial incident assessment. *Disaster Med Public Health Prep*. 2008;2(S1):S51–7.
19. Hung KKC, Mashino S, Chan EYY, Graham CA, Cowling BJ. Health workforce development in health emergency and disaster risk management: the need for evidence-based recommendations. *Int J Environ Res Public Health*. 2021;18(7):3382.
20. Jackson A. Should Australian disaster team deployments include a nurse leader? *Australas Emerg Nurs J*. 2011;14:57.
21. Jaiswal K, Wald D, D'Ayala D. Developing empirical collapse fragility functions for global building types. *Earthq Spectra*. 2011;27(3):775–95.
22. Joseph-Richard P. *Advanced Methods for Human Resource Management Research: Design, Implement and Evaluate a Research Project*. London: Kogan Page Publishers; 2025.
23. Kaptan K. An organizational metamodel for hospital emergency departments. *Disaster Med Public Health Prep*. 2014;8(5):436–44.
24. Kaya ŞŞ, Erdoğan EG. Being a nurse during an earthquake that affected ten provinces: a qualitative study on experiences and expectations. *Int Nurs Rev*. 2024;72(1):e13051.
25. Khorram-Manesh A. Flexible surge capacity: public health, public education, and disaster management. *Health Promot Perspect*. 2020;10(3):175–9.
26. Khoshmaram N, Gholipour K, Farahbakhsh M, Tabrizi JS. Strategies and challenges for maintaining the continuity of essential health services during a pandemic: a scoping review. *BMC Health Serv Res*. 2025;25(1):691.
27. Labrague LJ, Hammad K, Gloe DS, McEnroe-Petitte DM, Fronda DC, Obeidat AA, et al. Disaster preparedness among nurses: a systematic review of literature. *Int Nurs Rev*. 2018;65(1):41–53.
28. Liu Y, Wang H, Chen J, Zhang X, Yue X, Ke J, et al. Emergency management of nursing human resources and supplies to respond to coronavirus disease 2019 epidemic. *Int J Nurs Sci*. 2020;7(2):135–8.
29. Lokmic-Tomkins Z, Bhandari D, Bain C, Borda A, Kariotis TC, Reser D. Lessons learned from natural disasters around digital health technologies and delivering quality healthcare. *Int J Environ Res Public Health*. 2023;20(5):4542.
30. Merino Y, Ceferino L, Pizarro S, de la Llera JC. Modeling hospital resources based on global epidemiology after earthquake-related disasters. *Earthq Spectra*. 2024;40(4):2402–29.
31. Okur Ü, Özyayın T, Şen M. Afetlerde halk sağlığı hizmetlerine ilişkin hemşirelerin deneyimleri ve algıları: nitel bir çalışma. *Hacettepe Üniversitesi Hemşirelik Fakültesi Dergisi*. 2024;11(3):207–17.
32. Paul JA, George SK, Yi P, Lin L. Transient modeling in simulation of hospital operations for emergency response. *Prehosp Disaster Med*. 2006;21(4):223–36.
33. Pöler R, Mula J, Díaz-Madroño M. *Operations research problems statements and solutions*. Berlin: Springer; 2014.
34. Şenol Balaban M, Doğulu C, Akdede N, Akoğlu H, Karakayalı O, Yılmaz S, et al. Emergency response, and community impact after Feb 6, 2023 Kahramanmaraş Pazarcık and Elbistan earthquakes: reconnaissance findings and observations on affected region in Türkiye. *Bull Earthq Engineering*. 2025;23(3):1053–81.
35. Sever MS, Vanholder R. Management of crush-related injuries after disasters. *N Engl J Med*. 2012;367(19):1876–85. <https://doi.org/10.1056/NEJMr a1108940>.
36. Shiono K, Krimgold F, Ohta Y. Modeling of search-and-rescue activity in an earthquake. *Proceedings of the 10th WCEE, Balkema, Rotterdam*. 1992. 6043–6048.
37. Siddique H. GP appointments should be five minutes longer, says BMA. *The Guardian*. 2016.
38. So E. *Estimating fatality rates for earthquake loss models*. Berlin: Springer; 2016.
39. St Louis P, Wautié A, Makali SL, Radix C, Paul E. Can health system be resilient in a failed state? Lessons from a scoping review in Haiti. *Discov Health Systems*. 2025;4(1):74.
40. Taylor M, O'Neal JP, Nadeau K, Zerylnick J, Harris C. Enhancing resilience: a framework for increasing health care workforce knowledge about novel infectious diseases using Ebola Virus Disease as a case study. *Int J Disaster Risk Reduct*. 2025;120:105333.
41. TÜİK. 2023. Nüfus İstatistikleri [Population Statistics]. Retrieved March 10, 2025. <https://nip.tuik.gov.tr>.
42. USGS PAGER. 2023a. Retrieved March 10, 2025. <https://earthquake.usgs.gov/earthquakes/search/#%7B%22feed%22%3A%221437493916387%22%2C%22search%22%3A%7B%22id%22%3A%221437493916387%22%2C%22name%22%3A%22Search%20Results%22%2C%22isSearch%22%3Atrue%22%22params%22%3A%7B%22productype%22%3A%22losspager%22%2C>
43. USGS PAGER. 2023b. Retrieved March 10, 2025. <https://earthquake.usgs.gov/earthquakes/map/?extent=-89.34629,-298.125&extent=89.33821,658.125&range=search&format=losspager&listOnlyShown=true&timeZone=utc&search=%7B%22name%22%3A%22Search%20Results%22%22%22params%22%3A%7B%22starttime%22%3A%222023-02-05%2000:00>
44. USGS PAGER. 2023c. Retrieved March 10, 2025. <https://earthquake.usgs.gov/earthquakes/eventpage/us6000jllz/pager>.
45. WHO. 2019. *World Health Organization Health emergency and disaster risk management framework*. Geneva: World Health Organization. Licence: CC BY-NC-SA 3.0 IGO. ISBN 978-92-4-151618-1. Retrieved March 10, 2025. <https://iris.who.int/server/api/core/bitstreams/219a1a08-9ec5-4f2d-9aff-54c713fca7c/content>
46. Wu M-H, Yu J-Y, Huang C-H. Theoretical system dynamics modeling for Taiwan pediatric workforce in an era of national health insurance and low birth rates. *Pediatr Neonatol*. 2013;54(6):389–96.
47. Yardım M, Eser E. Ayaktan tanı ve tedavi başvurularında hasta başına kaç dakika ayrılmalıdır. *Turkish J Public Health*. 2017;15(1):58–67.
48. Yıldırım C, Kara M, Kaya D. Disaster triage outcomes after the 2023 Türkiye earthquakes: Lessons from emergency departments. *Turkish J Emerg Med*. 2023;23(3):117–23. https://doi.org/10.4103/TJEM.TJEM_52_23.
49. Zarea K, Beiranvand S, Sheini-Jaberi P, Nikbakht-Nasrabadi A. Disaster nursing in Iran: challenges and opportunities. *Australas Emerg Nurs J*. 2014;17(4):190–6.
50. Zhong L, Lopez D, Pei S, Gao J. Healthcare system resilience and adaptability to pandemic disruptions in the United States. *Nat Med*. 2024;30(8):2311–9.

Publisher's Note

Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.